MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BSM201 Mathematics - IIA

UPID : 002005

Time Allotted : 3 Hours *Full Marks : 70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) The median of the call received on 7 consecutive days 11,13,17,13,23,25,19_____is.

(ii) Regression equation of X on Y is used for estimating value of variable _____ for a given value of _____.

(iii) A statement made about a population for testing purpose is called _____.

(iv) Mutually Exclusive events does not contain any _____ sample point.

(v) The p.d.f. of exponential distribution is given by _____.

(vi) The random variables X and Y have variances 0.2 and 0.5 respectively and their covariance is zero. Let $Z = 5X - 2Y$. Then variance of Z is _____.

(vii) If X and Y are independent the s.d $(X - Y)$ = _____.

(viii) If the regression coefficient of X on Y and Y on X are -0.5 and -0.5 respectively then the correlation coefficient between X and Y is _____.

(ix) If the Critical region is evenly distributed then the test is referred as _____.

(x) The expected value of a discrete random variable 'x_i' with frequency f_i is given by _____.

(xi) If X is a normally-distributed random variable with a mean of 100 and a standard deviation 15 , then $P(X<100) =$

(xii) If the joint pdf of X and Y is given by $f(x,y) = e^{-(x+y)}, x>0, y>0$, then $E(XY) =$ _____.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. The joint pdf of a bivariate (X,Y) is $f(x,y) = c(x+y), x>0, y>0, x+y<2$. Find c and $P(X<1, Y>0.5)$. [5]

3. Find the mean of the following data: [5]

Daily wages (Rs) :	25–29	30–34	35–39	40–44
No. of workers :	16	28	14	12

4. In a bolt factory, machines A, B and C manufacture respectively 25%, 35% and 40% of the total output. 5%, 4% and 2% are defective bolts. A bolt is drawn at random from the product and found to be defective. What are the probabilities that it was manufactured by machines A, B and C? [5]

5. A salesman is expected to give an average sale of Rs. 3500 per day. Observing the sales of a particular salesman for 6 days, it is seen that he gives an average sale of Rs. 3300 per day with s.d. 1016.53. Using [5]

5% level of significance, test whether his work is below standard. Given $t_{0.05} = 2.02$ for 5 d.o.f.

6. The joint pdf of a bivariate (X,Y) is $f(x,y) = 2$, $x>0$, $y>0$. Find the conditional density of Y given X. Also find $P(Y>1/2 \mid X = 1/3)$. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) The median and mode of the following frequency distribution are known to be 27 and 26 respectively. Find the values of α and β. [8]

class interval :	0-10	10-20	20-30	30-40	40-50
frequency :	3	α	20	12	β

(b) Find the s.d. of the following frequency distribution: [7]

Weight (in kg) :	36-40	41-45	46-50	51-55	56-60	61-65	66-70
No of persons :	14	26	40	33	50	37	25

8. (a) Let [8]

$$F(x) = \begin{cases} 0, & x<0 \\ \frac{1}{5}, & 0\le x<1 \\ \frac{3}{5}, & 1\le x<3 \\ 1, & 3\le x \end{cases}$$

Show that $F(x)$ is a possible distribution function. Find the spectrum and the probability mass of the distribution. Also find the mean and standard deviation of X.

(b) A and B throw alternatively a pair of dice. A wins if he throws 8 before B throws 5, and B wins if he throws 5 before A throws 8. Find the probability that A wins if he starts the game. [7]

9. (a) Prove that $r_{xy} = 1$ if and only if y is a linear function of x. [10]

(b) If $Var(x) = 9, Var(y) = 4$ and $Var(x-y) = Var(x)$, then find the correlation coefficient between x and y. [5]

10. (a) A car-hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distribution with average number of demand per day 1.5. Calculate the proportion of days on which neither car is used and the proportion of days on which some demands are refused. Given $e^{-1.5} = 0.2231$. [5]

(b) In a certain factory producing razor blades there is a small chance 1/500 for a blade to be defective. The blades are in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing (i) no defective (ii) 1 defective (iii) 2 defective blades in one consignment of 10000 packets. Given $e^{-0.02} = 0.9802$. [6]

(c) If a random variable has a Poisson distribution such that P(1) = P(2), then find (i) mean (ii) s.d. (iii) P(4). [4]

11. Prove that $-1 \le \rho_{xy} \le 1$. [15]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : ESCS201 Programming for Problem Solving

UPID : 002008

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) Point out the difference between strcmp() and strncmp()

(ii) What do you mean by pointer arithmetic in C?

(iii) Write the main of the differences between Structure and Union?

(iv) Find the correct output:

```
#include<stdio.h>
main()
{
for(int i=0; i<=5;i++);
{
printf("%d",i);
}
}
```

(v) What is the main difference between call by value and call by address in C?

(vi) Why is scope of variable necessary in function?

(vii)
```
#include
main()
{
int i=3;
i=i++;
printf("%d",i);
}
```
What will be the output of the above code?

(viii) Why do we use header files?

(ix) Write the output of the following Code:

```
#include
int main()
{
char x;
x = 'a';
printf("%d\n", x);
}
```

(x) Define recursion.

(xi) Arrangement of playing cards is an example of which sorting technique?

(xii) Explain Ackerman function.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Differentiate between the switch and nested-if statement in C with proper example. [5]
3. Write a C program to count the number of even and odd numbers from an 1-D array. [5]
4. Write a program in C to find the area and circumference of a circle using function [5]
5. Write a C program to find the length of an array using pointer arithmetic. [5]
6. Write a C program to perform bit wise left shift operation on a number taken from the user. Write the output with an example. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Draw a flowchart to evaluate factorial of a number. [5]
   (b) Write a C program to generate Fibonacci Series upto n-terms using loop. [5]
   (c) Explain preprocessor directive with suitable example. [5]
8. (a) Define a recursive function to calculate factorial of a number and call this function to calculate $S = 1! + 2! + ...n!$, where n is user input. [8]
   (b) Write a C program to swap the content of two variables using Call by Value. [7]
9. (a) Write a C program to list all the prime numbers for a given range of numbers. [8]
   (b) Write a C program to check the entered character is a consonant or vowel. If it is not an alphabet return 'wrong input'. [7]
10. (a) Describe string with example. [5]
    (b) How strings are represented in memory? [4]
    (c) Write a C program to check the entered string is a palindrome or not. [6]
11. (a) Create a recursive C function to find the GCD of two numbers. [5]
    (b) Write a C program to sort names of students in a class. [10]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BSCH201 Chemistry-I (Gr-A)

UPID : 002002

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]
   (i) Write the reagents of nitration of benzene.
   (ii) Write the value of quantum numbers n, l and m for 2s orbital.
   (iii) MRI is the application of which spectroscopic method?
   (iv) London forces or dispersion forces operates between _____.
   (v) How the entropy of the system changes when water is frozen?
   (vi) What is Hard-soft acid base principle?
   (vii) If two stereoisomers of tartaric acid have 2S, 3R and 2S, 3S configurations, then they are _____.
   (viii) What are the conditions of Cannizzaro reaction?
   (ix) State whether naphthalene is aromatic, antiaromatic or nonaromatic and give reason.
   (x) Write the equation of rotational constant B of a molecule.
   (xi) What is the value of critical coefficient?
   (xii) Write the half-cell representation of saturated calomel electrode.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. What is the role of Lewis acid in halogenation of benzene? Write the structure of the substrate which on ozonolysis will provide acetone as the only product. [5]
3. Write the significance of van-der Waal's constants. [5]
4. What are conformational isomers? Explain by taking ethane as an example. Draw the potential energy diagram of ethane. [5]
5. Prove that for a constant pressure process, where work in only mechanical, the heat absorbed by the system (Q_P) is equal to the increase in enthalpy (ΔH). [5]
6. NO is paramagnetic while NO+ is diamagnetic. Justify using MO diagram. Write electronic configuration of NO. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Define 'Standard cell' with example. [3]
   (b) Find the relation between ΔH and ΔU for an ideal gas undergoing a P-V type mechanical work as follows: [4]
   $\Delta H = \Delta U + \Delta nRT$
   (c) Discuss any one statement of First law of thermodynamics and explain it mathematically. [4]
   (d) Explain 'Galvanic cell corrosion'. [4]
8. (a) Demonstrate the shape of the following compounds. [8]
   (i) ClF_3 (ii) XeF_2 (iii) BrF_5 (iv) SCl_6
   (b) Arrange NH_3, H_2O, CH_4 according to increasing bond angle stating the reason. [4]
   (c) Dipole moment of BF_3 is zero while that of NF_3 is 0.24D. Justify. [3]
9. (a) Derive Schrodinger Equation. [5]
   (b) Write the conditions of acceptable wave function Ψ. [5]
   (c) Write short note on extrinsic semiconductor and its types. [5]
10. (a) What will be the product/ products if $CH_3CH=CH_2$ is reacted with HBr? Show the steps of reaction. [5]

(b) What change will be observed in the above reaction if peroxide is added along with HBr? [2]

(c) Choose the more suitable substrate of E1 reaction between tertiary-butyl chloride and methyl chloride. Give reason. [3]

(d) Explain the stereo chemical aspects of SN^1 and SN^2 reactions with suitable examples. [5]

11. (a) Calculate the CFSE and Magnetic moment of $K_3[FeF_6]$. [5]

(b) Calculate the CFSE of both high spin and low spin complexes of d^7 and d^4 complexes. [5]

(c) Give the molecular energy level diagram of CO. Comment on magnetic behaviour and bond order. [5]

*** END OF PAPER ***

MAKAUT 1059

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : HMHU201 English

UPID : 002011

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is a simple sentence?

(II) Identify whether the sentence is correct or incorrect:
"*The cat and the dog plays in the garden.*"

(III) Choose the more descriptive sentence:
a) "*The cake was good.*"
b) "*The cake was soft, rich, and filled with creamy chocolate.*"

(IV) What should the subject line in a business email be like?

(V) Which word is a back-formation: "donate" or "donation"?

(VI) What is a compound sentence?

(VII) Choose the correct verb:
"*Either my mother or my father (is/are) coming to the meeting.*"

(VIII) Identify the stronger verb in this sentence:
"*The river (flowed/moved) swiftly through the valley.*"

(IX) What is the function of the CC and BCC fields in an email?

(X) The Greek root "*phon*" means "*sound.*" Identify the word that contains this root:
a) Graphic
b) Telephone
c) Vision
d) Manual

(XI) Rewrite the following simple sentence as a complex sentence: "*She studied hard to pass the exam.*"

(XII) Fill in the blank with the correct verb:
The quality of these products _____ excellent. (*is/are*)

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. What is the purpose of a cover letter? List three key elements that should be included in it. [5]
3. Explain the types of clipping with examples for each. [5]
4. Rewrite the following sentences by changing the type of phrase or clause used: [5]
a. "*She left early because she was feeling sick.*" (Change the clause to a phrase.)
b. "*Being tired, he decided to rest.*" (Change the phrase to a clause.)
5. Identify the noun-pronoun agreement error and correct the sentence: [5]
"*If a person wants to be successful, they must work hard.*"
6. What is classification in writing? Explain its role with an example. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) What is a misplaced modifier? Explain with two examples and correct them. [5]
(b) Define definite and indefinite articles. Provide three examples for each. [5]
(c) Explain the importance of prepositions in sentence structure with five examples. [5]
8. What is the role of examples and evidence in writing? Explain with three examples. [15]
9. Essay Writing: [15]
Write a well-structured essay of 200-250 words on the topic:
"*The Impact of Artificial Intelligence on Society*"

Your essay should include:
Introduction – Define AI and its significance.
Body – Discuss at least three key impacts of AI with examples.
Conclusion – Summarize the main ideas and provide a final thought.

10. Discuss the role of acronyms in vocabulary building. Explain their significance with five examples, including their full forms and meanings. [15]

11. The following paragraph lacks cohesion. Rewrite it to improve the logical flow, clarity, and connection between ideas. [15]

Paragraph (Lacking Cohesion):

Technology is important in our daily lives. Many people use smartphones. They also like social media. Computers help in offices. Some people prefer working from home. The internet provides information. Online shopping is common. It is convenient. Communication is faster now. People can send messages quickly.

Instructions:

Use transition words to improve connections between sentences.
Use pronoun references to avoid unnecessary repetition.
Use parallel structure to make sentences flow better.
Ensure the paragraph has a clear central idea with supporting details.

*** END OF PAPER ***